

Thermal and thermo-catalytic degradation of high-density polyethylene waste

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Abstract

Thermal and thermo-catalytic cracking behaviour of a commercial high-density polyethylene (HDPE) waste was investigated in the temperature range of 400–450 °C in a laboratory scale pyrex batch reactor. An equilibrium fluid catalytic cracking catalyst (FCC), HZSM-5 and a clinoptilolite containing rhyolite tuff were used as catalysts. The presence of a catalyst significantly altered both the yield and composition of the products and differences were found between catalysts having different acidities, micro- and macropore surface areas. The yield of gases increased in the order: non-thermo-catalytic cracking < clinoptilolite < FCC < HZSM-5 catalysed cracking, while the yields of liquid products increased in the sequence of non-thermo-catalytic cracking < clinoptilolite < HZSM-5 < FCC catalysed cracking. The catalysts significantly affected the isomerization of the olefin double bond and of the carbon-chain of the cracked products, the olefin content of liquids did not differ considerably. The effect of the catalysts on the yields and structure of products were getting less significant with increasing cracking temperature. The range of the carbon atom number in liquids was C₅–C₂₈ in non-thermo-catalytic cases, whereas using catalysts it was C₅–C₂₅, but the temperature of cracking had a noticeable effect on the distribution of carbon atom. Each liquid fraction is of low sulphur content, which may be advantageous for application as fuel.

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1. Introduction

Nowadays the question of utilization of plastic wastes is in the focus of waste management. One reason of this fact – over the environmental aspects – is that these materials have high, so called hidden energy content, therefore significant energy could be gained with their reasonable recycling. In 2000 the plastic consumption was 45 Mt in Europe, and the consumption is growing annually by 4–8%. On the other hand the amount of waste from them was about 30 Mt [1,2]. Nearly 90% of plastic wastes are utilized for landfilling and incineration, but both applications can induce further

environmental problems, e.g., air pollution. Therefore, other methods are suggested for treating of waste polymers [3,4]. Polyolefins (polyethylene and polypropylene) amount almost to the half of plastic wastes. With chemical recycling these materials might be converted into gas, liquid and solid semi-products, which are more valuable than the wastes or the products of mechanical recycling.

The main routes of the researches are the conversion of polyolefins over different type of zeolite catalysts into liquid and wax-like solid hydrocarbons. The chemical structure of polymers and the degradation temperature considerably affect the yields and properties of reaction products. The increasing temperature of cracking increases the yields of both gas and liquid products, accompanied by increasing coke formation. The coking decreases the catalytic activity

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due to blocking of active sites. Mainly acidic clays and zeolites (ZSM-5, mordenite, Y-zeolite, β -zeolite) were used as catalyst [5–16].

Manos et al. investigated the degradation of high-density polyethylene (HDPE) over different catalysts [17]. They found that the sequence of formation of lighter hydrocarbons is US-Y < Y-zeolite < β -zeolite < mordenite < HZSM-5, and in parallel with these the olefin content of products increased. The catalysts also significantly affected the carbon number distribution of hydrocarbon products.

Takuma et al. used H-gallosilicate to the degradation of low-density polyethylene (LDPE) in the temperature range 375–500 °C [18]. H-gallosilicate proved to be an efficient catalyst for the formation of aromatics, e.g., benzene, toluene, xylene (BTX), the concentration of which increased with temperature (it was 58% at 500 °C). The authors observed cyclization and aromatization of hydrocarbons with carbon number more than six above 450 °C. Both equilibrium and fresh fluid catalytic cracking catalysts from refineries are applicable as catalysts for waste plastic degradation [19,20]. The use of a laboratory “riser” simulator, 500 °C cracking temperature and short residence time (12 s) might result in great yields of C₃–C₄ gases and gasoline with high aromatic content [20]. The efficiency of catalysts can considerably be modified by increasing the number of active sites [21,22].

In the present paper the possibility of chemical recycling of waste HDPE was investigated. The latest publications mostly reported the thermal and thermo-catalytic degradation of small amount (<10 g) of polymers generally followed with thermogravimetric method and their connected techniques, e.g. TG-MS. In most cases so called “virgin” polymers were degraded, and not a real waste (i.e. waste from agriculture or packaging, municipal plastic waste, etc.). For this purpose the thermal and thermo-catalytic cracking behaviour of greater amount of waste high-density polyethylene obtained from the packaging industry was investigated in a pyrex batch reactor. Further utilization of formed fractions was aimed as fuels. Therefore mild thermal and thermo-catalytic cracking was carried out in a temperature range of 400–450 °C to avoid the fraction of aromatics. Various catalysts were applied to increase the yields of lighter liquids as gasoline and kerosene. Because of economical considerations beside H-ZSM5 less expensive catalysts as equilibrium fluid catalytic cracking catalyst and clinoptilolite containing rhyolite tuff were also investigated. The effect of the cracking temperature and catalysts on the yields and structure of products are surveyed.

2. Experimental

2.1. Raw material and catalysts

Commercial HDPE motor oil flasks were crashed to 5–7 mm particles and used as raw material. The melt flow

Table 1
The main properties of catalysts

Properties	NCM	FCC	HZSM-5
Humidity, % (w/w)	7.5	3.4	3.5
Average grain size (μm)	32.3	59.7	12.4
Si/Al	5.67	16.4	28.10
BET surface area (m^2/g)	72.6	148.1	300.1
Micropores (m^2/g)	1.3	78.9	225.3
Surface area of pores, 1.7–300 nm (m^2/g)	29.6	83.6	152.1
Micropores (cm^3/g)	0.073	0.132	0.193
Surface area of pores, 1.7–300 nm (cm^3/g)	–	0.155	0.118

index (MFI) (at 190 °C with 2160 N load) was 0.538 g/10 min, ash content 1.14%, Ti and Ca contents 0.545 and 0.055%, respectively. The used raw material had 14.1 ppm sulphur content from additives of polymer (e.g. antioxidant, fire-retardant and antifuming additives).

An equilibrium fluid catalytic cracking catalyst (FCC) and a clinoptilolite containing rhyolite tuff (NCM) were used as catalysts. Commercial H-formed ZSM-5 catalyst was used for comparison. Table 1 shows the main properties of the applied catalysts. Macro- and microporous surface areas, Si/Al ratios which are proportional to acidities and cationic compositions are significantly different. The characteristic grain sizes decrease in sequence of FCC > NCM > HZSM-5. Each catalysts had a little cation content. NCM catalyst had K⁺ (0.091%), Mg²⁺ (0.017%), Ca²⁺ (0.062%), Fe²⁺ (0.046%), while the equilibrium FCC catalyst contained Na⁺ (0.302%), Ni²⁺ (0.006%), Cu²⁺ (0.001%), La²⁺ (0.114%), Ce²⁺ (0.047%), Nd²⁺ (0.008%), Pr²⁺ (0.019%) and the HZSM-5 catalyst Ca²⁺ (0.036%) and Na⁺ (0.082%) cations.

2.2. Cracking apparatus and the analyses of products

The cracking of wastes was carried out for 1 h in a 500 cm³ pyrex batch reactor at 400, 420 and 450 °C inserting 200 g of waste polymer, and in case of thermo-catalytic cracking 3 wt.% catalyst was added to the polymer. The temperature of melted polymer was measured with a thermometer. For ensuring an inert atmosphere nitrogen was introduced into the polymer with a flow rate of 7 dm³/h. Volatile products formed in the cracking reactions were separated to liquids and gases in a condenser, where liquid products were condensed at 20 °C. The gases formed in cracking reactions and the introduced nitrogen mixed the catalyst with molten reaction mixture.

The degradation of HDPE resulted in forming of gases, liquid products and residue. The composition of gases was measured with a Carlo Erba Vega Series GC 6000 gas-chromatograph (GC) provided with a 50 m × 0.32 mm fused silica column with Al₂O₃/KCl coating, under isothermal conditions.

A TRACE gas-chromatograph (GC) fitted with a 30 m × 0.32 mm Rtx-1[®] (Crossbond[®], 100% dimethyl polysiloxane) column was used to investigate the liquids. The GC was controlled by ChromCard software.

Table 2
The yields of cracking of HDPE waste

	400 °C				420 °C				450 °C			
	None	NCM	FCC	HZSM-5	None	NCM	FCC	HZSM-5	None	NCM	FCC	HZSM-5
Gas (G)	2.4	4.1	5.3	12.0	2.9	5.8	5.5	15.7	5.8	6.3	6.3	15.1
Liquid (L)	3.1	5.3	15.8	14.3	11.5	28.2	30.4	28.9	74.5	78.5	82.5	81.0
Residue (R)	93.5	90.6	78.9	73.7	85.6	66.0	64.1	55.4	19.7	15.2	11.2	3.9

G = 100 – (L + R).

The olefin double bond distribution was determined by infrared (IR) technique, with a SHIMADZU IR-470 type spectrometer (resolution: 2.7 cm^{-1} , illumination: Global light, monochromator: Littrow prism, detector: KRS 5 type detector with a pyrometer in crystal window vacuum cell) in the $400\text{--}1200\text{ cm}^{-1}$ wavenumber range. The vinyl double bond gave two intensive absorption bands at 910 and 990 cm^{-1} , the vinylidene one at 890 cm^{-1} wavenumber and the internal double bond one at 956 cm^{-1} .

The density of liquids was measured with an automatic Anton-Paar oscillometric density-meter by the method of MSZ EN ISO 12185. The sulphur content of the liquid fraction was measured with a PHILIPS PW 4025/02 X-ray spectrometer in helium atmosphere.

Scanning electron microscope (SEM) provided with EDAX was used for analysing the catalyst surface before and after the cracking process.

3. Results and discussion

3.1. Yields

Table 2 shows the yields of the cracking of HDPE without and with catalysts. It was found that both gas and liquid yields increased with the increasing temperature of degradation, and nearly complete cracking could be attained at 450 °C .

At lower temperatures the yields of both volatile and liquid products are higher when using catalysts, showing the higher reaction rate of catalytic reaction. It was found that

the greater microporous surface area of HZSM-5 catalyst caused much higher catalyst activity than found for FCC and NCM catalysts having lower microporous surface areas. Primary cracking reactions of polymer chain may proceed on the macroporous surface of catalyst, while the smaller fragments are cracked on the microporous surface of the catalyst. These data can be correlated with the result of Park et al., who investigated the catalytic degradation of polypropylene in the temperature range of $350\text{--}450\text{ °C}$ [11].

The separated catalyst was then investigated using SEM with EDAX. Coke was found on the surface of the used catalysts. Fig. 1a and b show the surface of clinoptilolite catalyst before and after waste polymer cracking.

The waste polymer contained 0.54% Ti as TiO_2 pigment, therefore during cracking TiO_2 also deposited on the catalyst surface. In Fig. 1b it can be seen that the catalyst became more compact, with larger grain size in the thermo-catalytic cracking process.

The coke formation is responsible for catalyst deactivation it being more pronounced at higher temperature. Thus at 450 °C the difference between non-catalytic and thermo-catalytic cases was less than at 400 °C . At a temperature of 450 °C coking of the catalyst was higher, and this could block the active sites of the catalyst, thus the activity of the catalyst considerably decreased. On the other hand perhaps the reaction rate of thermal cracking increased quicker with temperature, than that of thermo-catalytic case. Carbonaceous deposits may be formed by recombination of cracking fragments and not from aromatics since aromatics were not found in liquid products. Moreover, their formation is

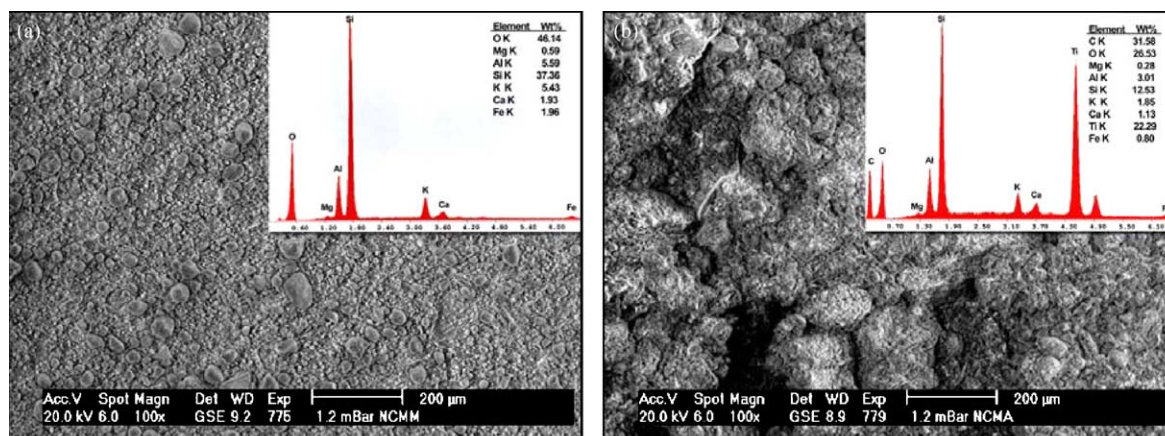


Fig. 1. The SEM micrograph and EDAX result of clinoptilolite catalyst: (a) before cracking and (b) after cracking process (450 °C).

Table 3

The composition of gases obtained by the cracking of HDPE waste (at 20 min of cracking time)

	400 °C				420 °C				450 °C			
	None	NCM	FCC	HZSM-5	None	NCM	FCC	HZSM-5	None	NCM	FCC	HZSM-5
C ₁	8.2	8.1	9.6	12.0	8.1	8.7	8.5	11.3	8	9.1	10.0	12.4
C ₂ =	11.2	10.4	11.7	9.2	11.4	11.4	9.0	9.0	11.4	11.0	10.3	12.1
C ₂	15.4	13.0	13.1	12.4	13.8	12.1	13.8	14.2	15.9	12.3	16.1	12.4
C ₃ =	11.0	13.3	14.1	14.1	10.9	13.7	13.6	12.4	11.1	13.7	16.0	14.2
C ₃	19.8	19.7	19.9	19.4	20.5	19.1	18.9	19.0	18.2	19.5	18.8	18.8
C ₄ =	11.6	10.9	10.4	10.9	12.2	11.8	12	12.7	13.3	13.1	9.1	9.2
C ₄	22.8	22.5	20.2	17.7	23.1	20.6	22	18.6	22.1	20.1	17.9	19.2
<i>i</i> -C ₄	0.0	2.1	2.0	4.3	0.0	2.6	2.2	2.8	0.0	1.2	1.8	1.7

improbable below 450 °C. Unfortunately simultaneously proceeding catalytic and thermal cracking cannot be separated. However, the product yields in function of temperature support a decrease in catalytic activity.

The residues of the samples cracked at 400 °C were hard and brittle, but became softer and more grease-like at 420 and 450 °C. This tendency was less pronounced in the presence of catalysts.

3.2. Composition of gases

The composition of non-condensed, i.e., at 20 °C volatile products, formed in cracking reactions is given in Table 3. C₂ and C₄ fractions are dominating. It is consistent with experimental data of other researchers [6,16,17].

In accordance with literature [12] the concentration of alkanes was higher than that of alkenes. However, the composition of gases did not change considerably with temperature or catalysts, though *i*-C₄ appeared only in the presence of catalysts. The concentration of *i*-C₄ was the highest in cracking of HDPE over HZSM-5 at 400 °C (4.3%), and it was the lowest over NCM at 450 °C (1.2%). The formation of methane and propane + propene was a bit higher with than without catalyst. All these results are in accordance with those found in yields and probably caused by the deactivation of catalysts.

3.3. Liquids

The composition of liquid products was determined by gas-chromatography. It was found, that liquids contained mostly C₅–C₂₈ aliphatic hydrocarbons. The carbon-chain distribution was considerably affected by catalysts. In the chromatograms mostly doublet peaks were observed. The first ones belong to *n*-alkenes and the second ones to *n*-alkanes. In thermo-catalytic cracking a great number of additional small peaks also appear which can be attributed to branched isomers (Fig. 2a and b).

Fig. 3a–c represent the carbon atom number distribution of liquid hydrocarbon components formed in cracking reactions. It is discernible that the range of the carbon atom number of liquid fraction became narrower by using catalysts and low cracking temperatures. For example, in case of non-catalytic cracking the number of carbon atoms in liquids ranges from C₅ to C₂₅ and this range was reduced by catalysts to C₅–C₂₂ at the same cracking temperature. Similar results have been published by Ding et al. [23], who investigated the thermal and catalytic degradation of polyethylene. On the basis of their experiments the range of the hydrocarbon products was C₁–C₂₇ in non-catalytic case, and C₁–C₁₇ in catalytic case [13].

Further utilization of liquids in the form of fuels seems to be a suitable way to improve the economy of recycling

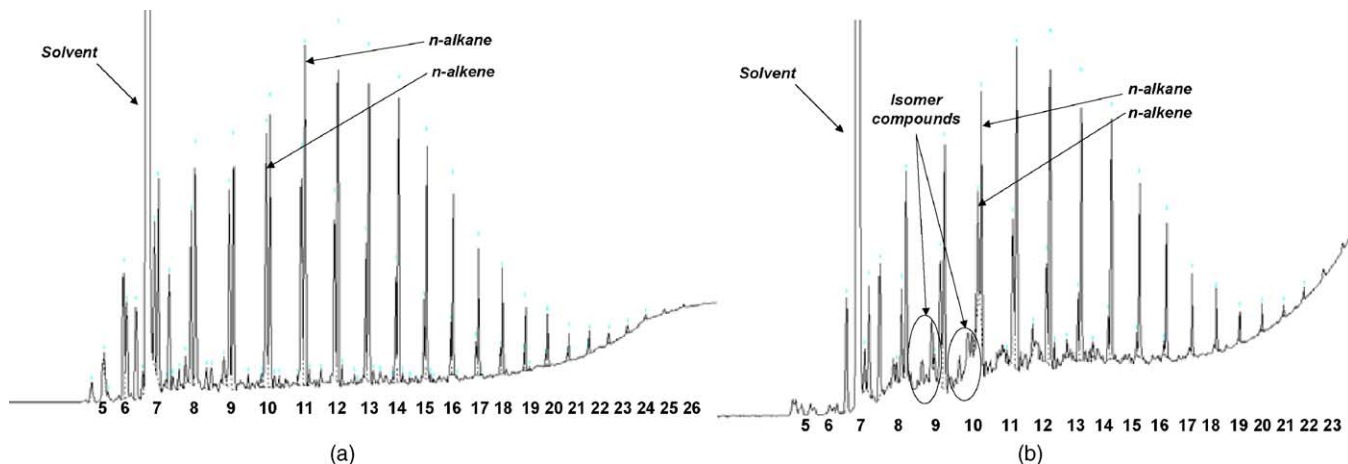


Fig. 2. The gas-chromatogram of liquid produced at 420 °C: (a) non-catalytic cracking and (b) over FCC.

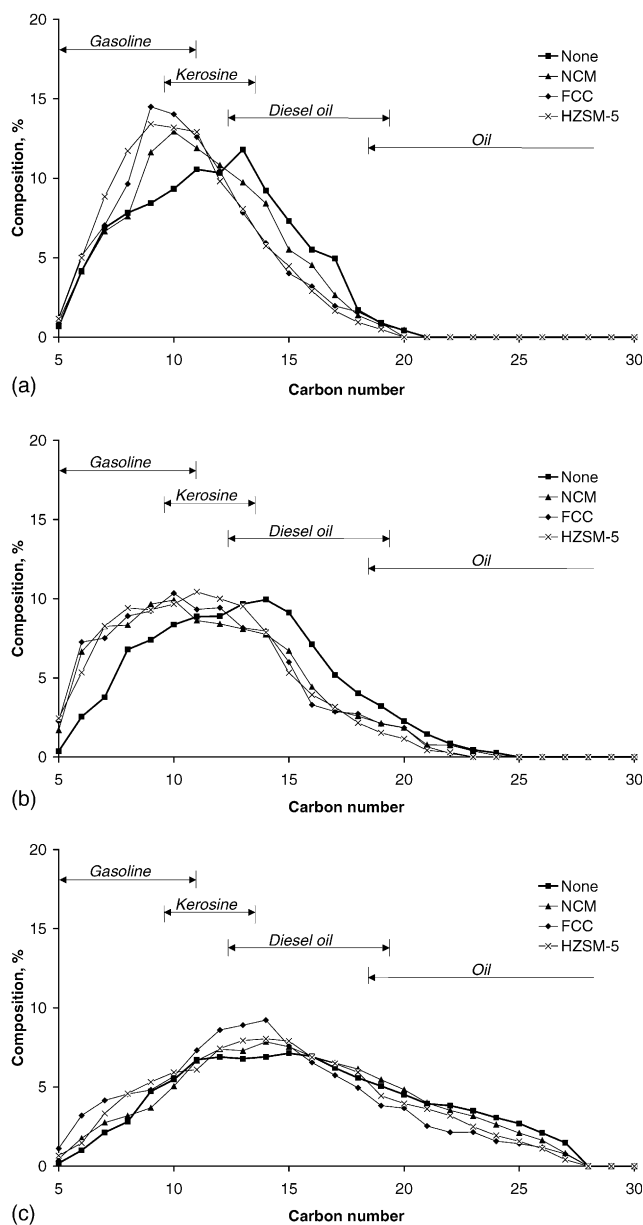


Fig. 3. Distribution of liquid fractions as a function of number of carbon atoms, produced at (a) 400 °C, (b) 420 °C and (c) 450 °C.

technology. As Fig. 3a–c show the main types of fuels are the gasoline and kerosene at lower cracking temperature, diesel oil and heavy oil at higher temperature (e.g. it was 25% – Fig. 4c – at 450 °C in non-catalytic case).

By using catalysts the concentration of lighter (C_{13} –) hydrocarbons increased in all cases, while the concentration of heavier (C_{13} +) alkene and alkane compounds decreased.

This effect can be attributed to the cracking activity of the catalysts. The differences between the activity of catalysts in the cracking process at different temperatures are not negligible. The difference between non-catalytic and thermo-catalytic cracking became less significant with increasing temperature in this aspect, too. It could be again the

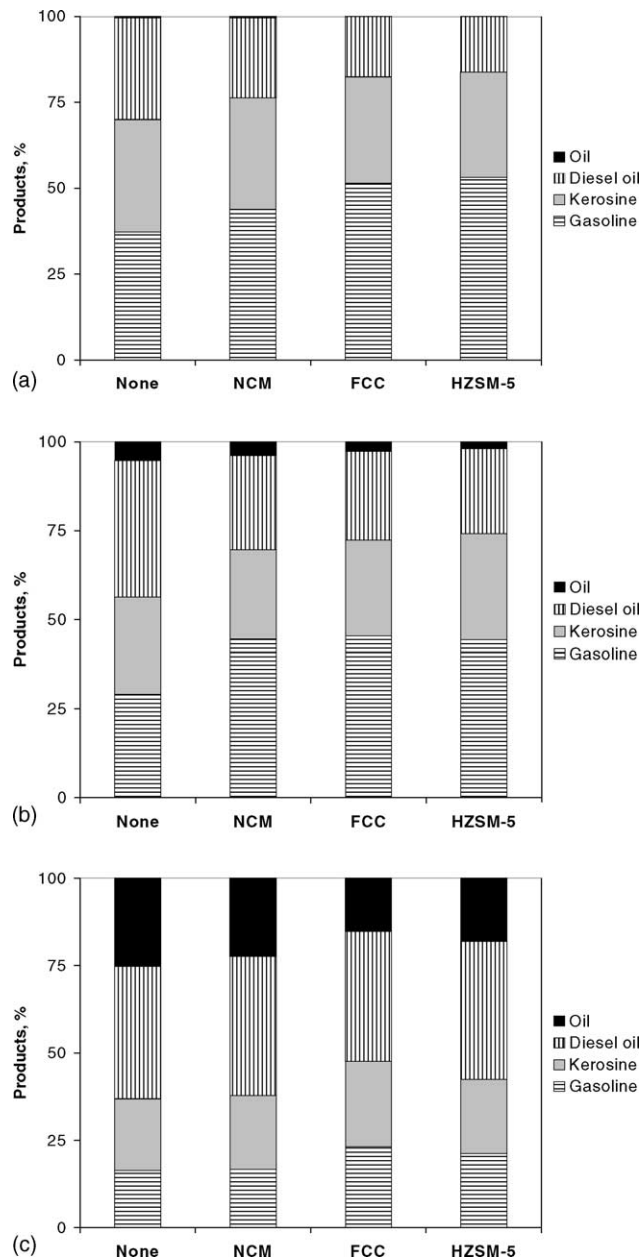


Fig. 4. The possibilities of further utilization of liquids produced at (a) 400 °C, (b) 420 °C and (c) 450 °C.

consequence of the coking of the catalyst surfaces at higher temperature, which may block the active sites of catalysts. The reaction rate of the thermo-catalytic cracking increases to less extent with increasing temperature than the reaction rate of the thermal cracking. According to visual observation the liquid was darker yellowish with increasing temperature of cracking. At higher temperature higher density of liquids was observed (Fig. 5) in each case of cracking. On the other hand the liquid density could be decreased with catalysts, mostly with HZSM-5 and equilibrium FCC.

It is well known that the thermal (non-catalytic) cracking of the plastics takes place by radical mechanism when the reaction is initiated by thermally generated radicals. The catalytic cracking on acid catalysts is initiated by formation

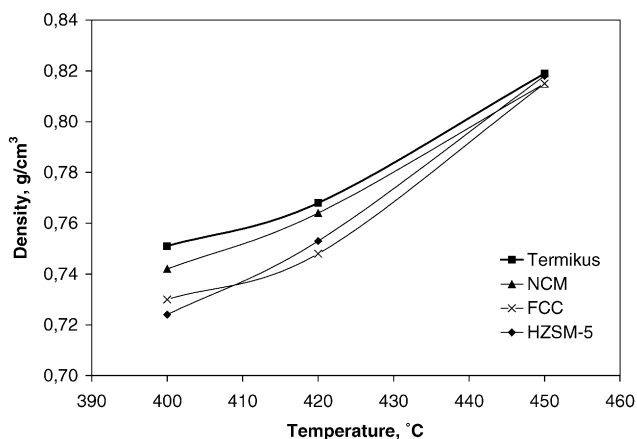


Fig. 5. The density of liquids at 20 °C.

of carbenium ions either by abstraction of a hydride ion from the polymer or by addition of a proton to the olefinic double bond of the polymer. The cracking of polymer carbocation proceeds thereafter step-by-step into lower molecular weight hydrocarbons.

Fig. 6a–c display the distribution of branched hydrocarbons as functions of carbon atoms. It is well discernible

that carbon-chain isomerization takes place in presence of catalysts, only. Yields of branched hydrocarbons decreased with increasing temperature. The branched hydrocarbons ranged between C₈–C₁₃ contributing to gasoline of high octane number. Carbon-chain isomerization activity of HZSM-5 catalyst was more significant in the carbon atom range of C₆–C₁₀ at 400 °C than that of the other catalysts. This difference can be attributed to the higher micro- and macroporous surface area of the H-ZSM5 catalyst (Table 1) but the difference diminished with increasing temperature. This also indicates that catalytic reactions are more and more restricted as coke formation deactivates the catalyst. The term “thermo-catalytic cracking” is used for describing this phenomenon.

The distribution of *n*-alkenes and *n*-alkanes as functions of number of carbon atoms are shown in the parts a–c of Figs. 7 and 8. Each distribution curve shows displacement to the lighter hydrocarbons. The distribution maxima belong, however, to different carbon atoms. The distribution curves are broader and the maxima are at higher carbon atom numbers in case of saturated compounds than in the case of unsaturated compounds. The possible reason of this phenomenon in case of larger hydrocarbons could be the termination reactions. In case of thermo-catalytic cracking it

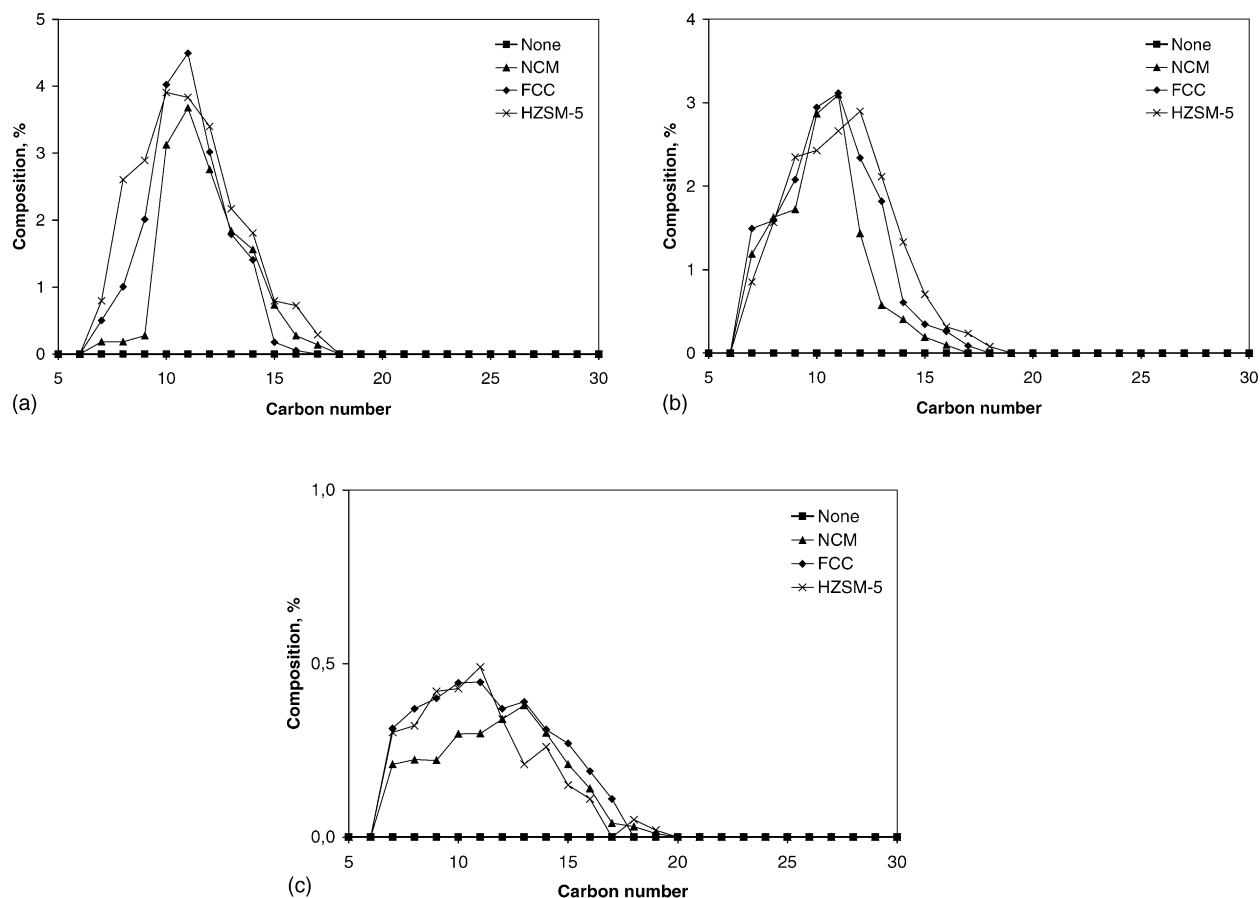


Fig. 6. The distribution of carbon atoms of branched hydrocarbons in liquids produced at (a) 400 °C, (b) 420 °C and (c) 450 °C.

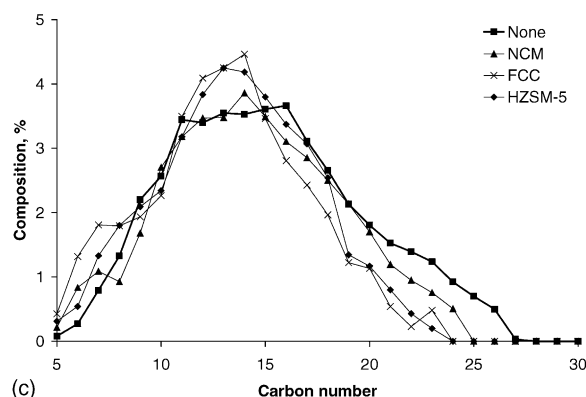
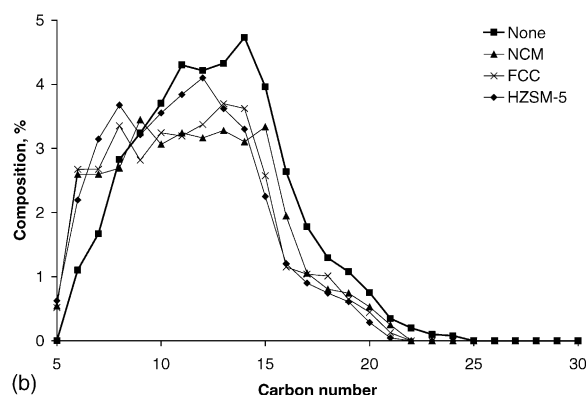
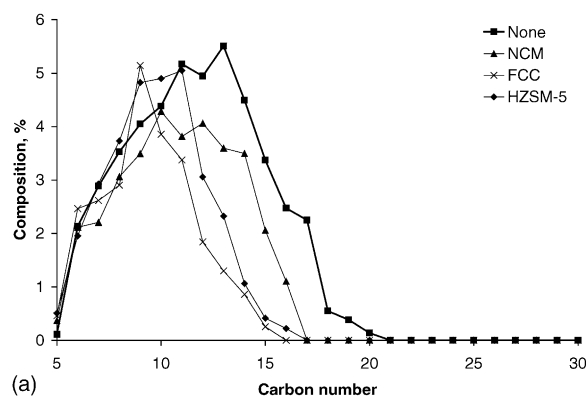


Fig. 7. The distribution of *n*-alkenes produced at (a) 400 °C, (b) 420 °C and (c) 450 °C.

was rather the recombination of carbenium ions producing alkanes. The possibility of hydride ion abstraction, which still resulted in alkenes was lower.

The olefin content of liquid fractions was investigated by infrared technique in the range of 1200–400 cm^{-1} wave-number. Fig. 9 shows the distribution of the different types of olefin double bond. The α -olefins, i.e. vinyl were the dominating olefins in thermal cracking while on catalysts this terminal double bond is shifted to internal position. In addition, isomerization of carbon-chain is observed on H-ZSM5 and FCC catalysts as indicated by the increase of vinylidene double bond content, but not with clinoptilolite. Though data of yields and gas composition suggested that

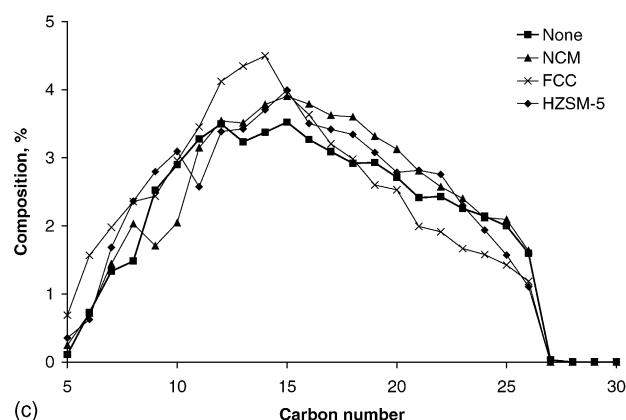
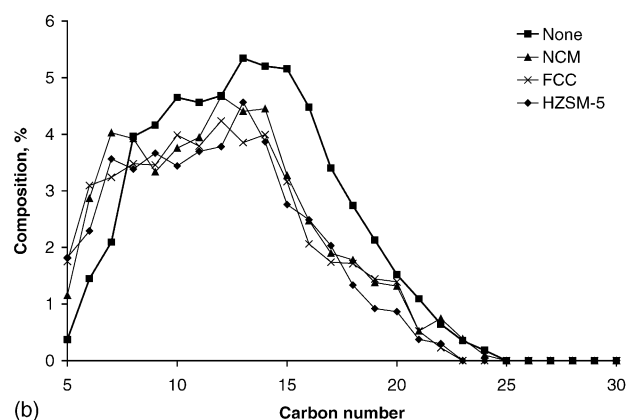
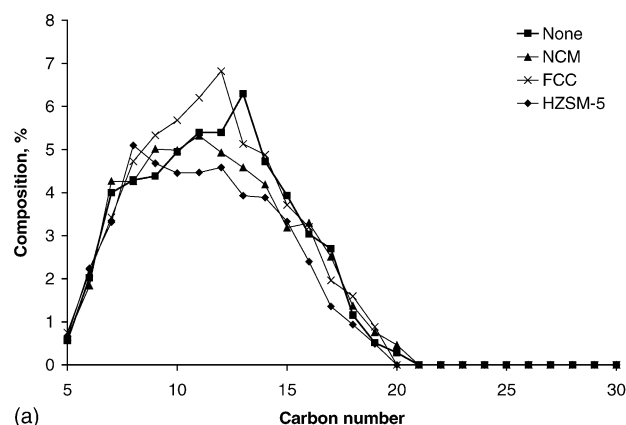


Fig. 8. The distribution of *n*-alkanes produced at (a) 400 °C, (b) 420 °C and (c) 450 °C.

the extent of the catalytic cracking reactions is fairly low, the effect of catalyst on the double bond distribution in liquid products is quite striking and does not depend on temperature.

The sulphur content of liquids is always an important parameter for their further utilization, because the low sulphur content might be one of their advantages in fuel-like utilization. The low concentration of sulphur of products is a favourable property both from environmental and humanbiological aspects. The sulphur content of

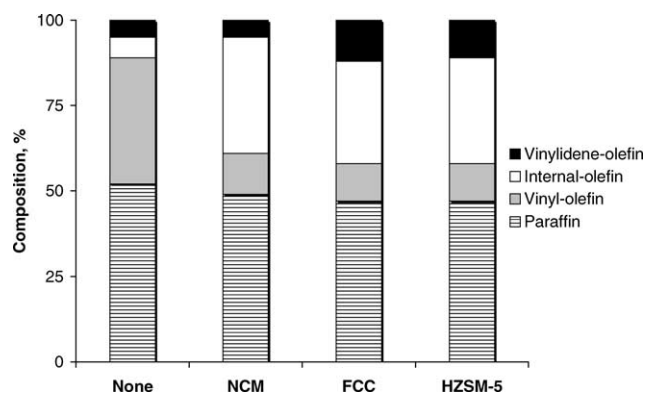


Fig. 9. The composition of liquids (cracking at 450 °C).

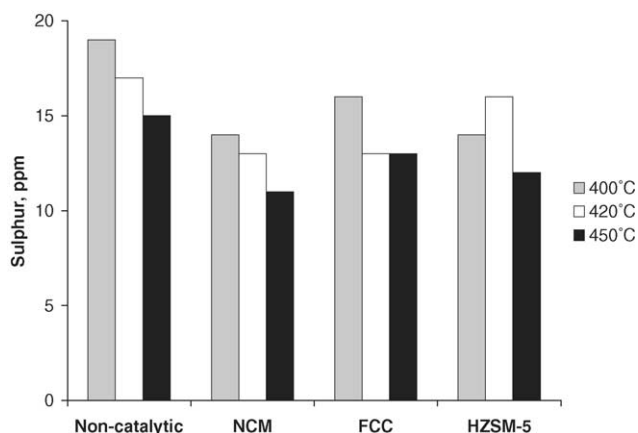


Fig. 10. The sulphur content of liquid cracked products.

liquids was determined by energy dispersive X-ray fluorescent technique. Fig. 10 represents the sulphur content of liquids. Each liquid fraction was of low (<20 ppm) sulphur content, which decreased by the use of catalysts. It was supposed that the catalysts adsorbed the sulphur content of liquids.

4. Conclusions

Thermo-catalytic and non-catalytic degradation of high-density polyethylene (HDPE) waste was investigated in a temperature range of 400–450 °C in pyrex batch reactor. Considerable differences were observed between yields, and composition of products. Both the increase of temperature and the use of catalysts increased the yield of products. At increasing temperature these differences decreased. Composition of gases containing mostly C₂ and C₄ hydrocarbons was not changed significantly with cracking parameters. Using catalysts the carbon number distribution of liquids became narrower, while the yields of gasoline and kerosene increased. Furthermore the catalysts enhanced significantly carbon-chain and double bond isomerizations. The catalytic effect decreased in the order

of HZSM-5 > FCC > clinoptilolite. The activity of catalysts decreased with cracking time, and the highest decrease was observed in case of HZSM-5, which had the highest initial activity. The sulphur content of each fraction was less than 20 ppm. These properties can be favourable for further fuel-like utilization.

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